Warehouse and Retail Sales Business Analysis

Standalone business review of demand patterns, channel efficiency, product performance, supplier concentration, transfer inefficiency, and optimization opportunities.

Total Sales	Warehouse Share	Retail Share	Top Category
9,942,656	**78.3%**	**21.7%**	**BEER**

Top 5 Supplier Share	Top 10 Product Share	Warehouse-Heavy Items	High-Transfer Items
59.7%	**16.7%**	**13,823**	**3,094**

Executive Summary

- Total Sales reconcile to **9,942,656** using the definition **Retail Sales + Warehouse Sales**.
- The business is warehouse-led, with warehouse contributing **78.3%** of total sales.
- **BEER** contributes **71.4%** of total sales.
- The top 5 suppliers drive **59.7%** of sales.
- 13,823 items are warehouse-heavy and 3,094 are high-transfer.

Business Implications

- Inventory planning appears optimized more for movement into channel than final sell-through.
- Beer, wine, and liquor should not share a single replenishment strategy.
- Supplier and SKU concentration make forecasting discipline disproportionately important.
- Long-tail SKUs likely tie up working capital and create avoidable transfer noise.

Data Quality & Caveats

All sales tables use Total Sales = Retail Sales + Warehouse Sales. Retail Transfers are excluded from Total Sales. Negative values were retained because they likely reflect returns, reversals, or operational corrections. Ratios are suppressed when total sales are negative or below 5.0.

Data Quality Summary

total_rows	total_columns	duplicate_rows	missing_supplier	missing_item_type	missing_retail_sales	negative_retail_sales	ne
307645	9	0	167	1	3	113	71

Coverage by Year

YEAR	min_month	max_month	months_present	rows
2017	6	12	7	96284
2018	1	2	2	26445
2019	1	11	11	138638

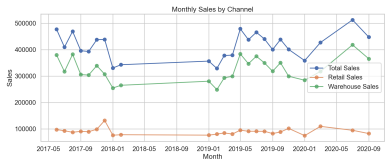
YEAR	min_month	max_month	months_present	rows
2020	1	9	4	46278

Methodology Box

Metric	Definition	Threshold / Note
Total Sales	retail_sales + warehouse_sales	Retail transfers excluded
Retail Share	retail_sales / total_sales	Suppressed when total_sales <= 5.0 or negative
Warehouse Share	warehouse_sales / total_sales	Suppressed when total_sales <= 5.0 or negative
Transfer-to-Sales Ratio	retail_transfers / total_sales	High-transfer if ratio > 75%; suppressed when denominator too small or negative
Warehouse-Heavy Item	warehouse_share > threshold	Threshold = 80%
Channel Imbalance	retail_share < low threshold or > high threshold	Low < 20%, high > 80%
Supplier Volatility (CV)	monthly_sales_std / avg_monthly_sales	Calculated at supplier-month level
Inefficiency Score	Weighted composite of low sales, transfer intensity, warehouse-heaviness, and non-positive months	Weights = 35%, 30%, 20%, 15%
Action Segments	Grow / Maintain / Reallocate/Monitor / Discount-Delist / Monitor	Grow<0.35, Maintain<0.45, Discount/Delist≥0.6

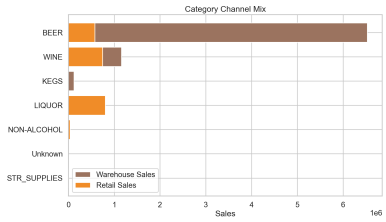
Key Graphics

Monthly Sales by Channel



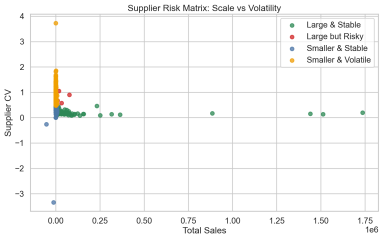
Warehouse movement dominates across periods, reinforcing a warehouse-led operating model.

Category Channel Mix



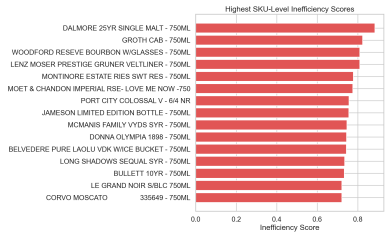
Channel balance differs sharply by category, so strategy should be category-specific.

Supplier Risk Matrix



Large volatile suppliers deserve tighter forecasting and contingency planning.

Highest SKU-Level Inefficiency Scores



The inefficiency list is deduplicated at full SKU grain to avoid repeated products.

Category Performance

Footnote: Total Sales = Retail Sales + Warehouse Sales; Retail Transfers excluded; negative values retained.

Category	Total Sales	Retail Sales	Warehouse Sales	Retail Transfers	Retail Share	Warehouse Share	Transfer-to-Sales Ratio	Unique Items
BEER	7,101,457.04	574,220.53	6,527,236.51	566,714.00	8.1%	91.9%	8.0%	5457
WINE	1,903,483.50	746,498.59	1,156,984.91	734,618.04	39.2%	60.8%	38.6%	21300
LIQUOR	897,597.70	802,691.43	94,906.27	794,735.71	89.4%	10.6%	88.5%	4475
KEGS	118,431.00	0.00	118,431.00	-1.00	0.0%	100.0%	-0.0%	2686
NON-ALCOHOL	60,233.90	34,084.31	26,149.59	26,666.38	56.6%	43.4%	44.3%	125
STR_SUPPLIES	2,740.88	2,740.88	0.00	10,846.58	100.0%	0.0%	395.7%	27
Unknown	1.00	0.00	1.00	0.00	0.0%	100.0%	0.0%	1
REF	-19,835.37	663.63	-20,499.00	388.92	-3.3%	103.3%	-2.0%	11
DUNNAGE	-121,454.00	0.00	-121,454.00	0.00	-0.0%	100.0%	-0.0%	4

Supplier Analysis

Supplier	Total Sales	Avg Monthly Sales	Monthly Sales Std Dev	Supplier CV	Retail Share	Supplier Segment
CROWN IMPORTS	1,736,309.17	72,346.22	14,526.70	20.1%	4.9%	Large & Stable
MILLER BREWING COMPANY	1,512,585.15	63,024.38	8,316.46	13.2%	5.8%	Large & Stable
ANHEUSER BUSCH INC	1,441,131.66	60,047.15	9,418.26	15.7%	7.6%	Large & Stable
HEINEKEN USA	885,936.37	36,914.02	6,329.24	17.1%	6.3%	Large & Stable
E & J GALLO WINERY	363,634.31	15,151.43	1,736.45	11.5%	45.7%	Large & Stable
DIAGEO NORTH AMERICA INC	315,907.83	13,162.83	1,798.51	13.7%	46.0%	Large & Stable
CONSTELLATION BRANDS	251,064.30	10,461.01	996.66	9.5%	52.4%	Large & Stable
BOSTON BEER CORPORATION	231,722.02	9,655.08	4,429.34	45.9%	17.3%	Large & Stable
YUENGLING BREWERY	157,175.56	6,548.98	932.99	14.2%	14.7%	Large & Stable
FLYING DOG BREWERY LLLP	154,651.04	6,443.79	909.99	14.1%	17.0%	Large & Stable

Supplier	Total Sales	Avg Monthly Sales	Monthly Sales Std Dev	Supplier CV	Retail Share	Supplier Segment
THE WINE GROUP	136,461.61	5,685.90	454.84	8.0%	47.1%	Large & Stable
PABST BREWING CO	124,152.22	5,173.01	818.01	15.8%	8.0%	Large & Stable
LEGENDS LTD	107,401.12	4,475.05	633.54	14.2%	8.6%	Large & Stable
JIM BEAM BRANDS CO	104,106.85	4,337.79	508.51	11.7%	92.4%	Large & Stable
REPUBLIC NATIONAL DISTRIBUTING CO	92,123.21	3,838.47	620.78	16.2%	27.8%	Large & Stable

Top Products

Item Code	Item Description	Supplier	Category	Total Sales	Retail Share	Warehouse Share
23445	CORONA EXTRA LOOSE NR - 12OZ	CROWN IMPORTS	BEER	328,224.83	7.6%	92.4%
96750	CORONA EXTRA 2/12 NR - 12OZ	CROWN IMPORTS	BEER	257,679.62	3.8%	96.2%
23886	HEINEKEN LOOSE NR - 12OZ	HEINEKEN USA	BEER	189,711.21	9.4%	90.6%
96970	HEINEKEN 2/12 NR - 12OZ	HEINEKEN USA	BEER	162,224.15	4.7%	95.3%
90590	MILLER LITE 30PK CAN - 12OZ	MILLER BREWING COMPANY	BEER	148,926.40	9.7%	90.3%
96741	CORONA EXTRA 4/6 NR - 12OZ	CROWN IMPORTS	BEER	132,665.61	5.7%	94.3%
25883	MODELO ESPECIAL 24 LOOSE NR - 12OZ	CROWN IMPORTS	BEER	121,846.80	4.0%	96.0%
90468	BUD LIGHT 30PK CAN	ANHEUSER BUSCH INC	BEER	108,615.97	11.3%	88.7%
96083	HEINEKEN 4/6 NR - 12OZ	HEINEKEN USA	BEER	105,184.92	5.5%	94.5%
23314	CORONA EXTRA 18PK NR - 12OZ	CROWN IMPORTS	BEER	100,664.44	0.4%	99.6%
99198	MODELO ESPECIAL 2/12 NR - 12OZ	CROWN IMPORTS	BEER	97,283.89	2.6%	97.4%
95800	MODELO ESPECIAL-2/12 CAN - 12OZ	CROWN IMPORTS	BEER	93,292.21	2.4%	97.6%
23777	MODELO ESPECIAL SUITCASE CANS - 12OZ	CROWN IMPORTS	BEER	93,072.92	1.6%	98.4%
91170	COORS LT 30PK CAN - 12OZ	MILLER BREWING COMPANY	BEER	86,834.74	9.3%	90.7%
97918	STELLA ARTOIS 2/12 NR - 11.2OZ	ANHEUSER BUSCH INC	BEER	76,791.69	10.6%	89.4%

Inventory & Transfer Inefficiency

Ranked at full SKU grain (`item_code + item_description + supplier + category`). Ratios suppressed when denominators are too small or negative.

Item Code	Item Description	Supplier	Category	Total Sales	Retail Transfers	Transfer-to-Sales Ratio	Warehouse Share	Inefficiency Score	Ratio Suppressed / Denominator Flag
74700	DALMORE 25YR SINGLE MALT - 750ML	E & J GALLO WINERY	LIQUOR	1.00	3.00			0.8848	True
235344	GROTH CAB - 750ML	GROTH VINEYARDS & WINERY	WINE	1.08	3.00			0.8237	True
72069	WOODFORD RESEVE BOURBON W/GLASSES - 750ML	BROWN-FORMAN BEVERAGES WORLDWIDE	LIQUOR	1.00	4.00			0.8098	True
82262	LENZ MOSER PRESTIGE GRUNER VELTLINER - 750ML	NICHE IMPORTERS	WINE	0.90	2.92			0.8093	True
353221	MONTINORE ESTATE RIES SWT RES - 750ML	DMV DISTRIBUTING LLC	WINE	2.25	9.00			0.7782	True
84578	MOET & CHANDON IMPERIAL RSE- LOVE ME NOW -750ML	MOET HENNESSY USA	WINE	0.17	72.00			0.7752	True
10299	PORT CITY COLOSSAL V - 6/4 NR	NORTH LOCK LLC DBA PORT CITY BREWING CO	BEER	0.17	3.00			0.7564	True
345455	JAMESON LIMITED EDITION BOTTLE - 750ML	PERNOD RICARD USA LLC	LIQUOR	0.80	3.00			0.7556	True
43041	MCMANIS FAMILY VYDS SYR - 750ML	REPUBLIC NATIONAL DISTRIBUTING CO	WINE	0.08	2.00			0.7460	True
317816	DONNA OLYMPIA 1898 - 750ML	REPUBLIC NATIONAL DISTRIBUTING CO	WINE	1.17	2.50			0.7449	True
166597	BELVEDERE PURE LAOLU VDK W/ICE BUCKET - 750ML	RELIABLE CHURCHILL LLLP	LIQUOR	0.17	7.00			0.7439	True
311161	LONG SHADOWS SEQUAL SYR - 750ML	REPUBLIC NATIONAL DISTRIBUTING CO	WINE	0.34	1.00			0.7352	True
319933	BULLETT 10YR - 750ML	RELIABLE CHURCHILL LLLP	LIQUOR	0.99	9.92			0.7330	True
320121	CORVO MOSCATO 335649 - 750ML	SOUTHERN GLAZERS WINE AND SPIRITS	WINE	0.08	0.25			0.7210	True

Item Code	Item Description	Supplier	Category	Total Sales	Retail Transfers	Transfer-to-Sales Ratio	Warehouse Share	Inefficiency Score	Ratio Suppressed / Denominator Flag
344992	LE GRAND NOIR S/BLC 750ML	UNITED STATES DISTILLED PRODUCTS	WINE	0.08	1.00			0.7210	True

Ratio Issue Summary

metric	value
raw_retail_share_gt_100pct	4
raw_transfer_to_sales_gt_100pct	1707
suppressed_ratio_rows	15487
duplicate_sku_rows_after_aggregation	0

Segmentation Summary

Performance Segment	Action Segment	Items	Total Sales	Avg Inefficiency
Low	Discount/Delist	172	-142,015.95	0.6766
Low	Grow	0	0.00	NaN
Low	Maintain	0	0.00	NaN
Low	Monitor	0	0.00	NaN
Low	Reallocate/Monitor	12108	18,647.06	0.4562
Medium	Discount/Delist	4	18.56	0.6156
Medium	Grow	0	0.00	NaN
Medium	Maintain	11843	114,566.78	0.3455
Medium	Monitor	5	40.10	0.5021
Medium	Reallocate/Monitor	160	819.38	0.4861
High	Discount/Delist	0	0.00	NaN
High	Grow	11474	9,949,398.32	0.2219
High	Maintain	0	0.00	NaN
High	Monitor	11	381.74	0.3980
High	Reallocate/Monitor	25	799.66	0.3819

Prioritized Recommendations

1. Rationalize long-tail SKUs using the corrected inefficiency ranking.
2. Tighten transfer governance for high-transfer items that are not translating into healthy sales outcomes.
3. Use category-specific channel strategies rather than a single replenishment logic.
4. Protect top suppliers and top SKUs with better forecasting and service-level management.

5. Separate operational categories like REF and DUNNAGE from commercial scorecards.